$$\frac{dx}{dp} = \frac{(p+5)-p}{(p+5)^2} = \frac{5}{(p+5)^2}$$

Elasticity of supply: $\eta_s = \frac{p}{x} \cdot \frac{dx}{dp}$

$$= (p+5)\frac{5}{(p+5)^2} = \frac{5}{p+5}$$

When $p = 20$, $\eta_s = \frac{5}{20+5} = 0.2$

Interpretation:

- If the price increases by 1% from $p = ₹ 20$, then the quantity of supply increases by 0.2% approximately.
- If the price decreases by 1% from $p = ₹ 20$, then the quantity of supply decreases by 0.2% approximately.

Example 6.12

For the cost function $C = 2x\left(\frac{x+5}{x+2}\right)+7$, prove that marginal cost (MC) falls continuously as the output x increases.

Solution:

$$C = 2x\left(\frac{x+5}{x+2}\right)+7 = \frac{2x^2+10x}{x+2}+7$$

Marginal cost:

$$MC = \frac{dC}{dx} = \frac{d}{dx}\left[\frac{2x^2+10x}{x+2}+7\right]$$

$$= \frac{(x+2)(4x+10)-(2x^2+10x)}{(x+2)^2}$$

$$= \frac{2(x^2+4x+10)}{(x+2)^2} = \frac{2\left[(x+2)^2+6\right]}{(x+2)^2}$$

$$= 2\left[1+\frac{6}{(x+2)^2}\right]$$

$\Rightarrow$ as x increases, MC decreases.

Hence proved.

Example 6.13

$\overline{C} = 0.05x^2 + 16 + \frac{100}{x}$ is the manufacturer's average cost function. What is the marginal cost when 50 units are produced and interpret your result.

Solution:

Total cost: $C = AC \times x = \overline{C} \times x$

$$= 0.05x^3 + 16x + 100$$

Marginal cost: $MC = \frac{dC}{dx} = 0.15x^2 + 16$

$$\left(\frac{dC}{dx}\right)_{x=50} = 0.15(50)^2 + 16$$

$$= 375 + 16 = ₹ 391$$

Interpretation:

If the production level is increased by one unit from $x = 50$, then the cost of additional unit is approximately equal to ₹ 391.

Example 6.14

For the function $y = x^3 + 19$, find the values of x when its marginal value is equal to 27.

Solution:

$y = x^3 + 19$

$\frac{dy}{dx} = 3x^2$ ... (1)

$\frac{dy}{dx} = 27$ ... (2) [Given]

From (1) and (2), we get

$3x^2 = 27 \Rightarrow x = \pm 3$

Example 6.15

The demand function for a commodity is $p = \frac{4}{x}$, where p is unit price. Find the instantaneous rate of change of demand with respect to price at $p = 4$. Also interpret your result.

06_11th_BM-STAT_Ch-6-EM.indd 137 21-04-2020 12:21:44 PM